3 - Sunny - 3N - 0Y
2 - Cloudy 2Y - 0N
3 - Rainy 2Y - 1N

2 - Hot - 1N, 1Y
3 - Mild - 1N, 2Y
3 - Cool - 1Y, 2N

	Outlook	Temper	Humidity	Windy	Rainy.
1)	Sunny	Hot	high	Weak	No
2)	Sunny	Mild	high	Strong	No
3)	Cloudy	Hot	high	Weak	Yes
4)	Rainy	Mild	high	Weak	Yes
5)	Rainy	Cool	Normal	Weak	Yes
6)	Rainy	Cool	Normal	Strong	No
7)	Cloudy	Mild	Normal	Strong	Yes
8)	Sunny	Cool	Normal	Weak	No

Entropy of Rainy:

$$= -\left(\frac{4}{8}\log_2\frac{4}{8}\right) - \left(\frac{4}{8}\log_2\frac{4}{8}\right)$$

$$= -(0.5\log_2 0.5) - (0.5\log_2 0.5)$$

$$= 1$$

Entropy of Att In Outlook:

Entropy of Sunny:

$$= -\left(\frac{0}{3}\log_2\frac{0}{3}\right) - \left(\frac{3}{3}\log_2\frac{3}{3}\right)$$

$$= -(0\log_2 0) - (1\log_2 1)$$

$$= 0$$

Entropy of Cloudy:

$$= -\left(\frac{2}{2}\log_2\frac{2}{2}\right) - \left(\frac{0}{2}\log_2\frac{0}{2}\right)$$

$$= 0$$

Entropy of Rainy:

$$= -\left(\frac{2}{3}\log_2\frac{2}{3}\right) - \left(\frac{1}{3}\log_2\frac{1}{3}\right)$$

$$= 0.918$$

CamScanner

4 - high - 2N, 2Y
4 - Normal 2Y, 2N

pg # 3

Entropy of Att in Humidity :

Entropy of High :- $\left(\frac{2}{4}\log_2\frac{2}{4}\right) - \left(\frac{2}{4}\log_2\frac{2}{4}\right)$

$= 1$

Entropy of Normal : $-\left(\frac{2}{4}\log_2\frac{2}{4}\right) - \left(\frac{2}{4}\log_2\frac{2}{4}\right)$

$= 1$

Info Gain (Hum) : $1 - \left(\frac{4}{8}\times 1 + \frac{4}{8}\times 1\right.$

$\boxed{IG\ (Hum) : 0}$

5 - weak 3Y-2N
3 - Strong 2N-1Y

Entropy of ~~Att~~ of Windy :

Entropy of weak :- $\left(\frac{3}{5}\log_2\frac{3}{5}\right) - \left(\frac{2}{5}\log_2\frac{2}{5}\right)$

$= 0.970$

Entropy of Strong :- $\left(\frac{1}{3}\log_2\frac{1}{3}\right) - \left(\frac{2}{3}\log_2\frac{2}{3}\right)$

$= 0.918$

$IG(Windy) = 1 - \left(\frac{5}{8}\times 0.970 + \frac{3}{8}\times 0.918\right)$

$= 1 - 0.9505$

$\boxed{IG\ Wind = 0.0495}$

Information Gain (OL) = $1 - \left(\frac{3}{8} \times 0 + \frac{2}{8} \times 0 + \frac{3}{3} \times 0.918\right)$

$= 1 - 0.918$

$IG\ (OL) = 0.082$

Entropy of Att in Temperature:

Entropy of Hot = $-\left(\frac{1}{2} \log_2 \frac{1}{2}\right) - \left(\frac{1}{2} \log_2 \frac{1}{2}\right)$

$= 1$

Entropy of Mild = $-\left(\frac{2}{3} \log_2 \frac{2}{3}\right) - \left(\frac{1}{3} \log_2 \frac{1}{3}\right)$

$= 0.918$

Entropy of Cool = $-\left(\frac{1}{3} \log_2 \frac{1}{3}\right) - \left(\frac{2}{3} \log_2 \frac{2}{3}\right)$

$= 0.918$

IG (Tem) = $1 - \left(\frac{2}{8} \times 1 + \frac{3}{3} \times 0.918 + \frac{3}{8} \times 0.918\right)$

$= 1 - 2.086$

$IG\ (Tem) = 1.086$

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Info Gain

0.082	Outlook
1.086	Temperature ✓
0	Humidity
0.0495	Windy

